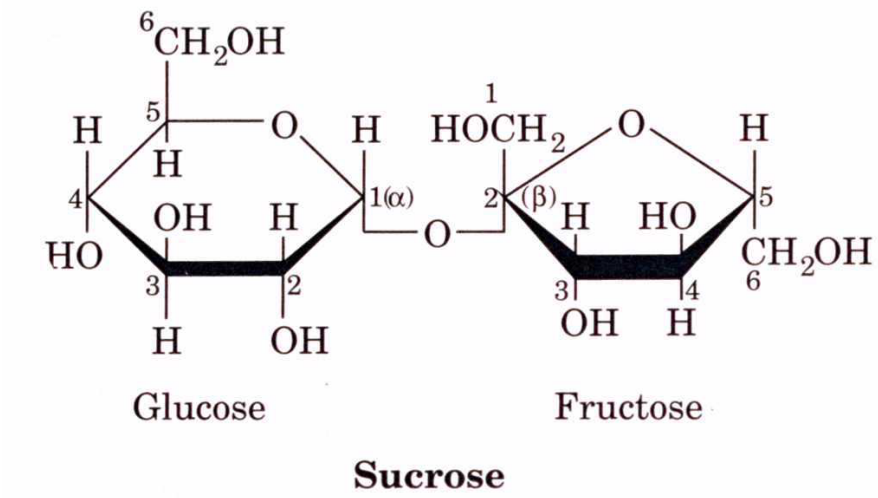
Enzyme Kinetics

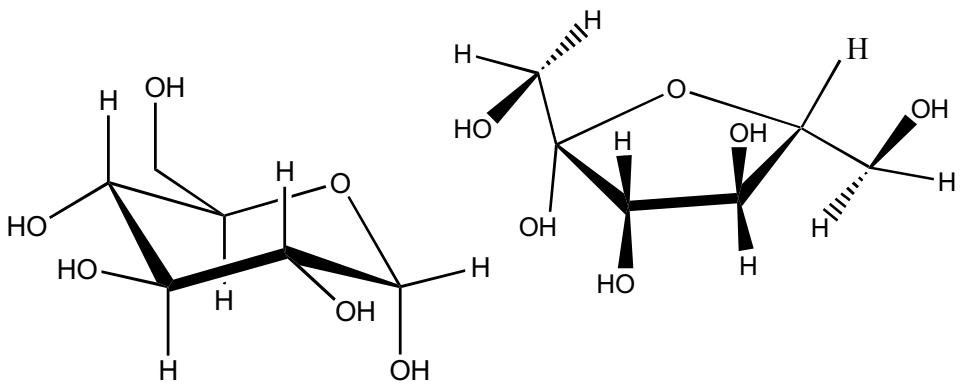
Kinetics of Enzymes

Enzymes follow zero order kinetics when substrate concentrations are high. Zero order means there is no increase in the rate of the reaction when more substrate is added.

Given the following breakdown of sucrose to glucose and fructose

Sucrose + H_20 ⟶ Glucose + Fructose





$$E + S \underset{k_{-1}}{\overset{k_1}{\leftrightarrow}} ES \xrightarrow{k_2} E + P$$

E = Enzyme S = Substrate P = Product

ES = Enzyme-Substrate complex

k_1 rate constant for the forward reaction

k_{-1} = rate constant for the breakdown of the ES to substrate

k_2 = rate constant for the formation of the products

When the substrate concentration becomes large enough to force the equilibrium to form completely all ES the second step in the reaction becomes rate limiting because no more ES can be made and the enzyme-substrate complex is at its maximum value.

$$v = \frac{d[\mathrm{P}]}{dt} = k_2[\mathrm{ES}]$$

[ES] is the difference between the rates of ES formation minus the rates of its disappearance.

1

$$\frac{d[\mathrm{ES}]}{dt} = k_1[\mathrm{E}][\mathrm{S}] - k_{-1}[\mathrm{ES}] - k_2[\mathrm{ES}]$$

Assumption of equilibrium

$k_{-1} >> k_2$ the formation of product is so much slower than the formation of the ES complex. That we can assume:

$$K_s = \frac{k_{-1}}{k_1} = \frac{[E][S]}{[ES]}$$

K_s or K_d is the dissociation constant for the ES complex.

$$[E]_T = [E] + [ES]$$

2

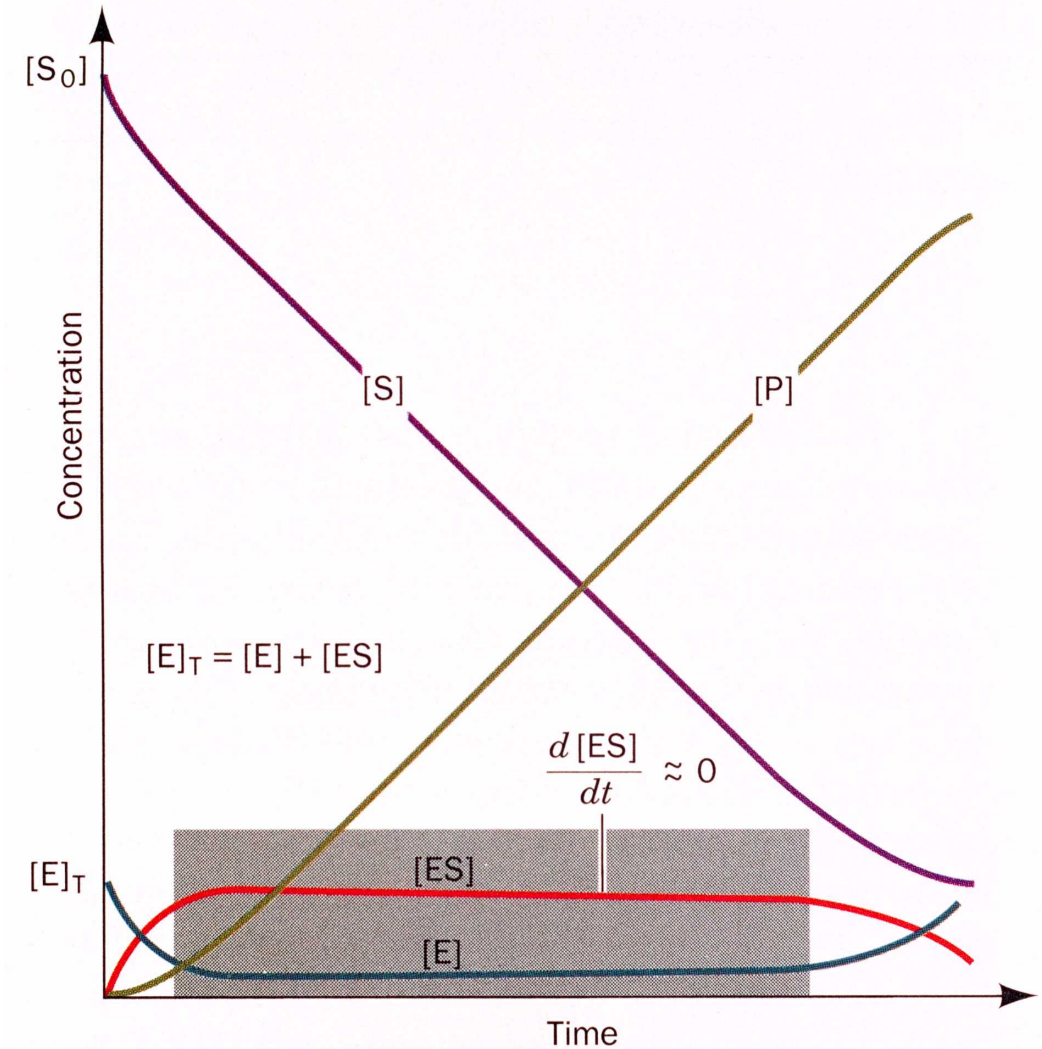
$$\mathbf{E} + \mathbf{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathbf{ES} \xrightarrow{k_2} \mathbf{E} + \mathbf{P} \qquad (6\text{–}10)$$

V_0 is determined by the breakdown of ES to form product, which is determined by [ES]:

$$V_0 = k_2[\mathbf{ES}] \qquad (6\text{–}11)$$

Assumption of steady state

Transient phase where in the course of a reaction the concentration of ES does not change



3

$$\frac{d[\mathrm{ES}]}{dt} = 0$$

Step 1 The rates of formation and breakdown of ES are determined by the steps governed by the rate constants k_1 (formation) and $k_{-1} + k_2$ (breakdown), according to the expressions

$$\textbf{Rate of ES formation} = k_1([\mathbf{E_t}] - [\mathbf{ES}])[\mathrm{S}] \tag{6–12}$$

$$\textbf{Rate of ES breakdown} = k_{-1}[\mathbf{ES}] + k_2[\mathbf{ES}] \tag{6–13}$$

Step 2 We now make an important assumption: that the initial rate of reaction reflects a steady state in which [ES] is constant—that is, the rate of formation of ES is equal to the rate of its breakdown. This is called the **steady-state assumption.** The expressions in Equations 6–12 and 6–13 can be equated for the steady state, giving

$$k_1([\mathbf{E_t}] - [\mathbf{ES}])[\mathrm{S}] = k_{-1}[\mathbf{ES}] + k_2[\mathbf{ES}] \tag{6–14}$$

Step 3 In a series of algebraic steps, we now solve Equation 6–14 for [ES]. First, the left side is multiplied out and the right side simplified to give

$$k_1[\mathbf{E_t}][\mathrm{S}] - k_1[\mathbf{ES}][\mathrm{S}] = (k_{-1} + k_2)[\mathbf{ES}] \tag{6–15}$$

Adding the term $k_1[\mathrm{ES}][\mathrm{S}]$ to both sides of the equation and simplifying gives

$$k_1[\mathbf{E_t}][\mathrm{S}] = (k_1[\mathrm{S}] + k_{-1} + k_2)[\mathbf{ES}] \tag{6–16}$$

We then solve this equation for [ES]:

$$[\mathbf{ES}] = \frac{k_1[\mathbf{E_t}][\mathrm{S}]}{k_1[\mathrm{S}] + k_{-1} + k_2} \tag{6–17}$$

This can now be simplified further, combining the rate constants into one expression:

$$[\mathrm{ES}] = \frac{[\mathrm{E_t}][\mathrm{S}]}{[\mathrm{S}] + (k_2 + k_{-1})/k_1} \tag{6–18}$$

The term $(k_2 + k_{-1})/k_1$ is defined as the Michaelis constant, K_m. Substituting this into Equation 6–18 simplifies the expression to

$$[\mathrm{ES}] = \frac{[\mathrm{E_t}][\mathrm{S}]}{K_\mathrm{m} + [\mathrm{S}]} \tag{6–19}$$

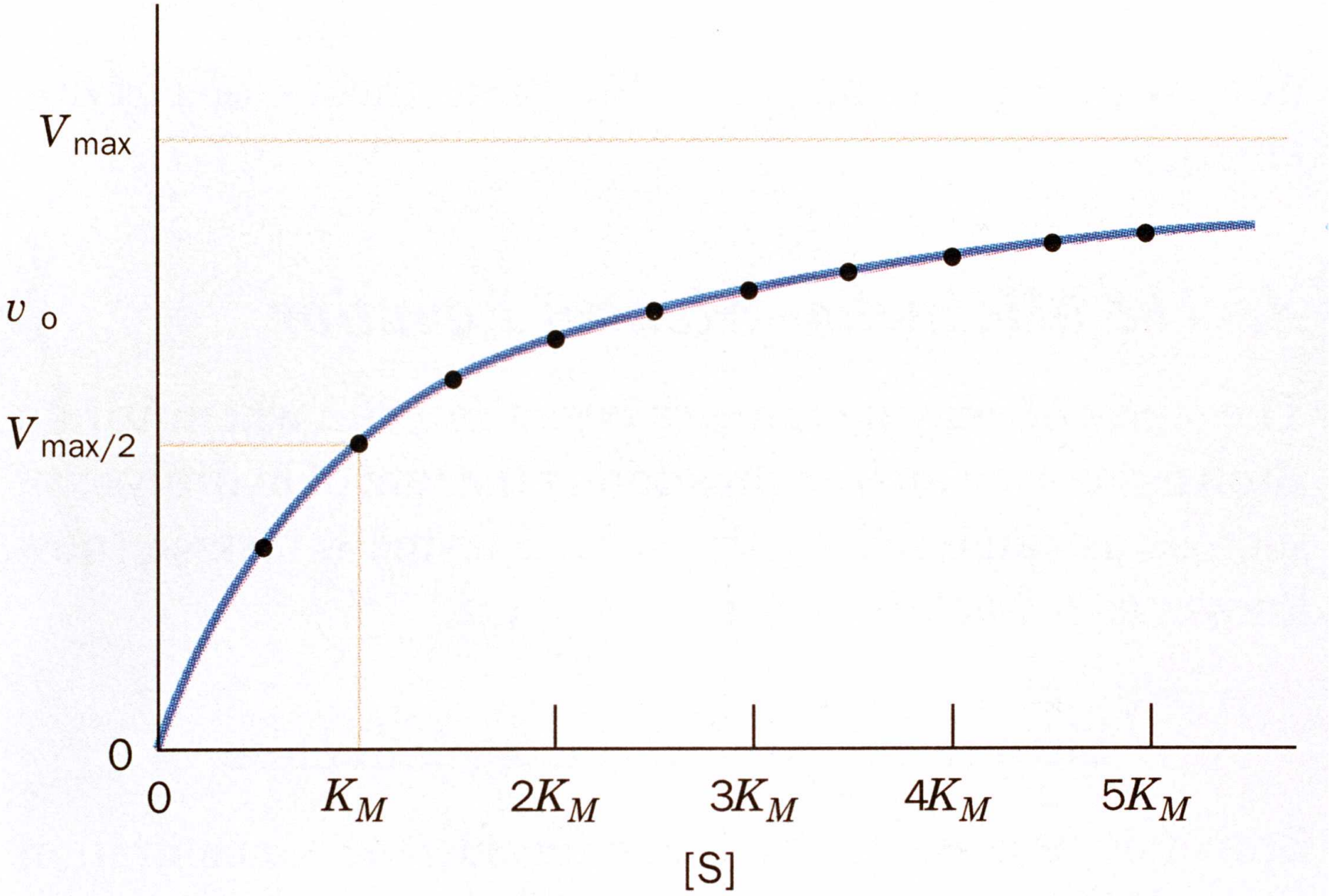
Step 4 We can now express V_0 in terms of [ES]. Substituting the right side of Equation 6–19 for [ES] in Equation 6–11 gives

$$V_0 = \frac{k_2[\mathrm{E_t}][\mathrm{S}]}{K_\mathrm{m} + [\mathrm{S}]} \tag{6–20}$$

This equation can be further simplified. Because the maximum velocity occurs when the enzyme is saturated (that is, with $[\mathrm{ES}] = [\mathrm{E_t}]$) V_{max} can be defined as $k_2[\mathrm{E_t}]$. Substituting this in Equation 6–20 gives Equation 6–9:

$$V_0 = \frac{V_{\mathbf{max}}\,[\mathrm{S}]}{K_{\mathbf{m}} + [\mathrm{S}]}$$

This is the **Michaelis-Menten equation,** the **rate equation** for a one-substrate enzyme-catalyzed reac-



The K_m is the substrate concentration where v_o equals one-half V_{max}

The K_M widely varies among different enzymes

The K_M

can be expressed as:

$$K_M = \frac{k_{-1}}{k_1} + \frac{k_2}{k_1} = K_s + \frac{k_2}{k_1}$$

As K_s or K_d decreases, the affinity for the substrate increases. The K_M can be a measure for substrate affinity if $\mathbf{k_2 < k_{-1}}$

TABLE 13-1. THE VALUES OF K_M, K_{CAT}, AND K_{CAT}/K_M FOR SOME ENZYMES AND SUBSTRATES

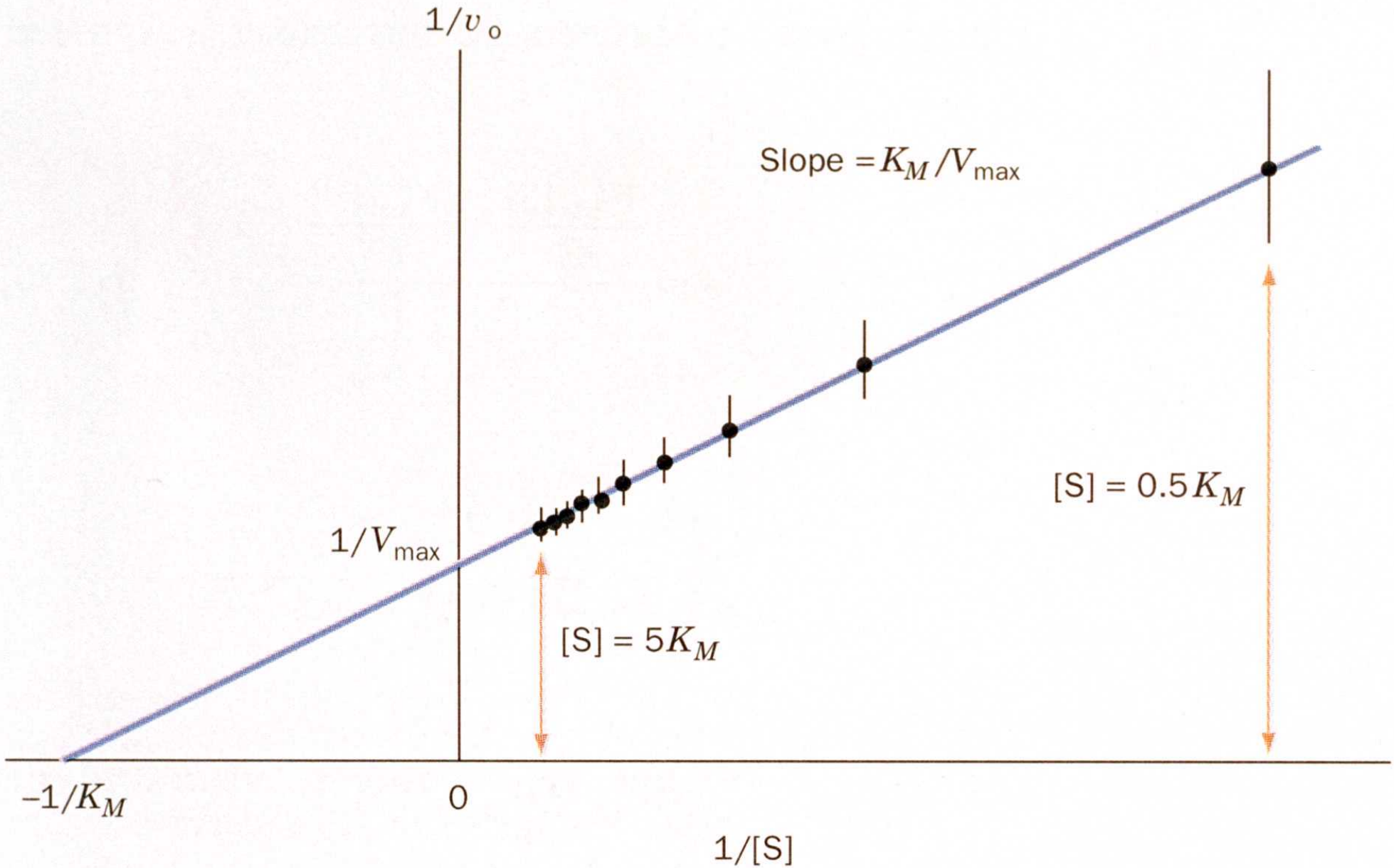
Enzyme	Substrate	K_M (M)	k_{cat} (s^{-1})	k_{cat}/K_M (M^{-1} s^{-1})
Acetylcholinesterase	Acetylcholine	9.5×10^{-5}	1.4×10^{4}	1.5×10^{8}
Carbonic anhydrase	CO_2	1.2×10^{-2}	1.0×10^{6}	8.3×10^{7}
	HCO_3^-	2.6×10^{-2}	4.0×10^{5}	1.5×10^{7}
Catalase	H_2O_2	2.5×10^{-2}	1.0×10^{7}	4.0×10^{8}
Chymotrypsin	*N*-Acetylglycine ethyl ester	4.4×10^{-1}	5.1×10^{-2}	1.2×10^{-1}
	N-Acetylvaline ethyl ester	8.8×10^{-2}	1.7×10^{-1}	1.9
	N-Acetyltyrosine ethyl ester	6.6×10^{-4}	1.9×10^{2}	2.9×10^{5}
Fumarase	Fumarate	5.0×10^{-6}	8.0×10^{2}	1.6×10^{8}
	Malate	2.5×10^{-5}	9.0×10^{2}	3.6×10^{7}
Urease	Urea	2.5×10^{-2}	1.0×10^{4}	4.0×10^{5}

There are a wide range of K_M, Vmax , and efficiency seen in enzymes

But how do we analyze kinetic data?

The double reciprocal plot

$$\frac{1}{v_o} = \left(\frac{K_M}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$


1/v o
Slope = K_M /V_max
[S] = 0.5 K_M
1/V_max
[S] = 5K_M
−1/K_M
0
1/[S]

Lineweaver-Burk plot: slope = K_M/Vmax,

$1/v_o$ intercept is equal to 1/Vmax

the extrapolated x intercept is equal to $-1/K_M$

For small errors in at low [S] leads to large errors in $1/v_o$

$$k_{cat} = \frac{V_{max}}{[E]_T}$$

k_{cat} is how many reactions an enzyme can catalyze per second

The turnover number

For Michaelis -Menton kinetics $k_2 = k_{cat}$

When $[S] << K_M$ very little ES is formed and $[E] = [E]_T$

and

$$v_o \approx \frac{k_2}{K_M}[E]_T[S] \approx \frac{k_{cat}}{K_M}[E][S]$$

K_{cat}/K_M is a measure of catalytic efficiency

The Eadie-Hofstee and Hanes plots

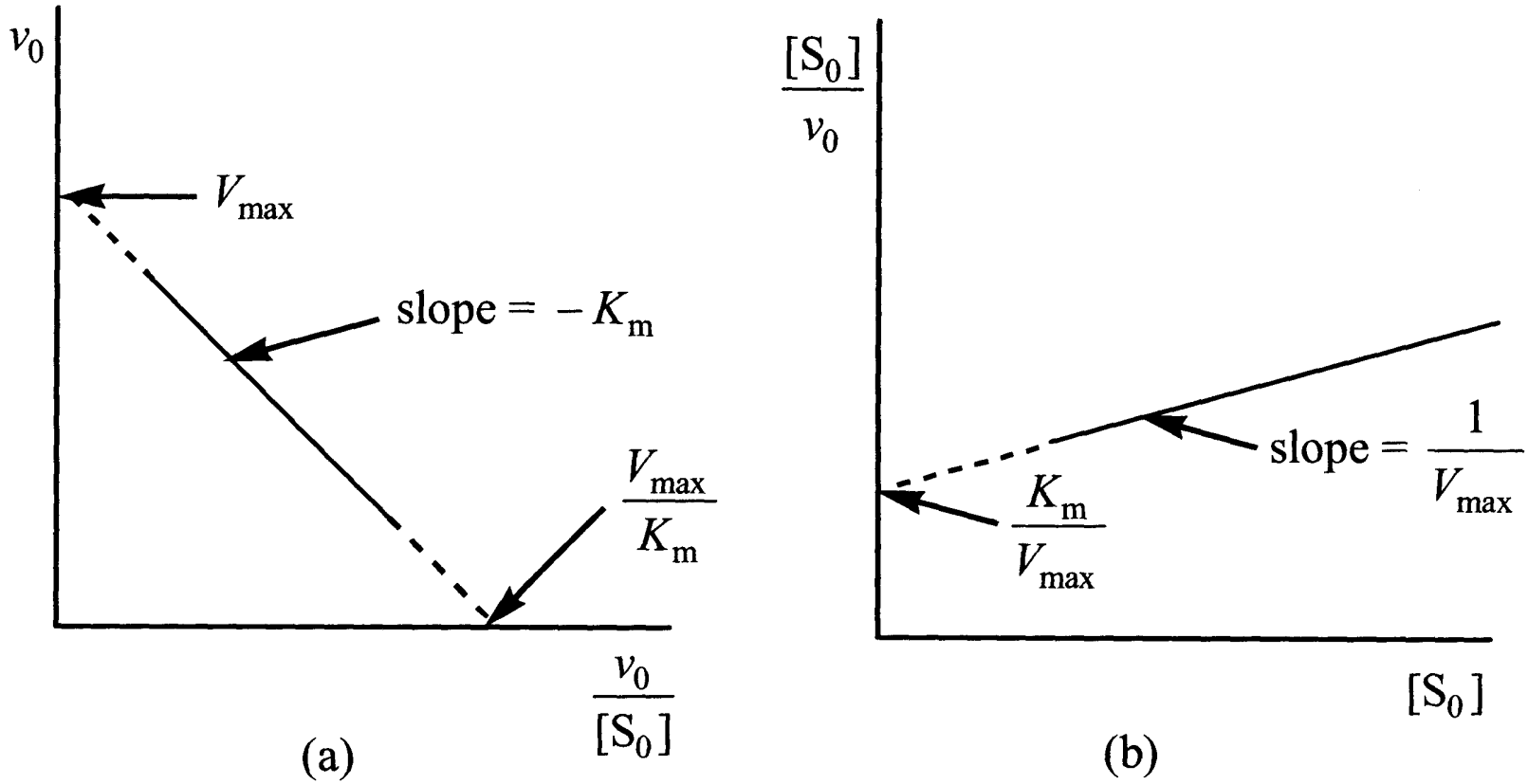
$$\frac{1}{(v_0)}(v_0)V_{max} = \frac{K_m}{(V_{max})}.\frac{1}{[S_0]}v_0(V_{max}) + \frac{1}{(V_{max})}v_0(V_{max}\}$$

$$\therefore\ v_0 = -K_m\frac{v_0}{[S_0]} + V_{max}$$

The **Hanes plot**

$$\frac{1}{v_0}[S_0] = \frac{K_m}{V_{max}}.\frac{1}{[S_0]}[S_0] + \frac{1}{V_{max}}[S_0]$$

$$\therefore\ \frac{[S_0]}{v_0} = \frac{1}{V_{max}}[S_0] + \frac{K_m}{V_{max}}$$



(a) The Eadie-Hofstee plot; (b) the Hanes plot.